

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO1

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of Artificial Intelligence

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the class of Artificial Intelligence that would surpass human ability at every task.

- A) Artificial Super Intelligence
- B) Artificial Narrow Intelligence
- C) Artificial General Intelligence
- D) Artificial Reactive Intelligence

Q2. Name the functionality class of Artificial Intelligence that keeps no memory of earlier inputs.

- A) Limited memory systems
- B) Reactive machines
- C) Theory of mind systems
- D) Self-aware systems

Q3. State the component of an intelligent agent that receives information from the environment.

- A) Its actuators
- B) Its performance measure
- C) Its sensors
- D) Its goal state

Q4. Identify the stage of the Artificial Intelligence development lifecycle in which the problem and its success criteria are agreed.

- A) Data preparation

- B) Model training
- C) Deployment
- D) Problem definition

Q5. Name the stage of the lifecycle in which the behaviour of a released system is tracked over time.

- A) Data collection
- B) Monitoring
- C) Feature engineering
- D) Model selection

Q6. Define the agent function.

- A) A mapping from every action to a numeric reward
- B) A mapping from every state to its utility value
- C) A mapping from every sensor to an actuator
- D) A mapping from every percept sequence to an action

Q7. Identify the type of environment in which the outcome of an action is not certain.

- A) Stochastic
- B) Deterministic
- C) Episodic
- D) Discrete

Q8. Name the agent type that maintains an internal model of the part of the world it cannot currently see.

- A) A simple reflex agent
- B) A table-driven agent
- C) A model-based reflex agent
- D) A utility-based agent

Q9. State the Responsible Artificial Intelligence principle that requires similar cases to be treated alike.

- A) Privacy
- B) Transparency
- C) Fairness
- D) Accountability

Q10. Identify the term for a confidently stated but false claim produced by a generative model.

- A) A hallucination
- B) An outlier
- C) A false negative
- D) An overfit

Q11. Name the field that draws on statistics, programming and domain knowledge to obtain insight from data.

- A) Deep learning
- B) Robotics
- C) Computer vision
- D) Data science

Q12. Identify the manufacturing application of Artificial Intelligence that anticipates when a machine will fail.

- A) Payroll processing
- B) Predictive maintenance
- C) Inventory labelling
- D) Document archiving

Q13. State the term for the instruction text supplied to a generative model.

- A) A percept
- B) A label
- C) A feature
- D) A prompt

Q14. Name the healthcare application of Artificial Intelligence that reads scans in support of a specialist.

- A) Appointment scheduling
- B) Payroll calculation
- C) Medical image analysis
- D) Stock ordering

Q15. Identify the environment property that holds when several agents are acting at the same time.

- A) Single-agent
- B) Multi-agent
- C) Static
- D) Episodic

Q16. Name the smart-city application of Artificial Intelligence that adjusts signal timings from live traffic.

- A) Adaptive traffic control
- B) Payroll automation
- C) Document translation
- D) Image compression

Q17. Explain why deep learning is described as a subset of machine learning.

- A) It was developed before machine learning existed
- B) It is one family of methods for learning from data
- C) It needs no training data of any kind at all
- D) It applies only to problems written out as rules

Q18. Distinguish a goal-based agent from a utility-based agent.

- A) A goal-based agent tests whether the goal is met, a utility-based one grades outcomes
- B) A utility-based agent tests whether the goal is met
- C) Both of the two only test whether the goal is met
- D) Both of the two grade every outcome numerically

Q19. Describe what it means for a task environment to be static.

- A) It contains exactly one agent at any moment
- B) It has a finite number of distinct states
- C) It gives the agent complete percepts always
- D) It does not change while the agent is deliberating

Q20. Interpret the following: an agent's sensors reveal only the room it currently occupies and nothing of the rest of the building.

- A) The environment is fully observable
- B) The environment is deterministic
- C) The environment is partially observable
- D) The environment is episodic

Q21. Predict what happens to a deployed model when the data it meets in service drifts away from its training data.

- A) Its accuracy falls although the code is unchanged
- B) Its accuracy improves as it sees more data
- C) It automatically retraines itself on the new data
- D) It refuses to produce any further predictions

Q22. Recognize the setting described: a robot arm learns to grasp an unfamiliar object by trying and receiving a score after each attempt.

- A) Supervised classification of grasps
- B) Unsupervised clustering of objects
- C) Reinforcement learning applied to robotics
- D) Rule-based expert reasoning

Q23. Explain how generative Artificial Intelligence differs from a classifier.

- A) It assigns a label rather than composing content
- B) It composes new content rather than assigning a label
- C) It requires far less training data than a classifier
- D) It cannot be applied to text of any kind

Q24. Summarize what the Artificial Intelligence development lifecycle covers.

- A) Only the training of the model on the data
- B) Only the writing of the software interface
- C) Only the collection and cleaning of the data
- D) Everything from defining the problem to monitoring the release

Q25. Distinguish a deterministic environment from a stochastic one.

- A) In a stochastic one the action fixes the next state
- B) Both of the two have next states fixed by the action
- C) In a deterministic one the action fixes the next state
- D) Neither of the two has any defined next state

Q26. Explain why a system that affects people must have an accountable owner.

- A) The software licence requires a named owner
- B) The model runs faster when an owner is named
- C) An owner improves the accuracy of the model
- D) Someone must answer for an outcome the system produces

Q27. Interpret a system that continues to make accurate predictions but is being used on a population quite unlike the one it was built for.

- A) Its reported accuracy is not evidence for this population
- B) Its accuracy will improve on the new population
- C) The population makes no difference to the model
- D) The model will refuse to run on the new data

Q28. Predict the risk of using a generative model to answer legal questions with no expert check.

- A) It will refuse to answer any legal question

- B) It may state a plausible but non-existent provision
- C) It will always cite the wrong section number
- D) It will answer far more slowly than usual

Q29. Describe what makes a task environment continuous rather than discrete.

- A) It contains an unlimited number of agents
- B) It never changes while the agent deliberates
- C) It gives the agent access to the whole state
- D) Its states and actions vary smoothly rather than in steps

Q30. Explain the difference between data science and machine learning.

- A) Machine learning is broader and includes data science
- B) Data science is broader and aims at insight for people
- C) The two terms describe exactly the same activity
- D) Data science never makes use of any data at all

Q31. Distinguish a learning agent from an agent whose rules are fixed.

- A) A fixed agent improves from the outcomes it observes
- B) Both of the two improve from what they observe
- C) A learning agent improves from the outcomes it observes
- D) Neither of the two responds to its current percept

Q32. Explain why Artificial Intelligence in healthcare is normally used to support rather than replace a clinician.

- A) The clinician remains accountable and sees context the model cannot
- B) The model is far slower than the clinician is
- C) The model cannot process any medical images
- D) Regulations forbid any use of AI in medicine

Q33. Recognize the ethical issue described: a model's training records contain far fewer examples from one section of the population.

- A) Representation bias in the training data
- B) Overfitting to the training data
- C) A leak between the training and test data
- D) An error in the choice of the loss function

Q34. Summarize the applications of Artificial Intelligence in robotics named on the syllabus.

- A) Payroll processing and document archiving
- B) File compression and network routing

- C) Spreadsheet calculation and report printing
- D) Perception, navigation and manipulation by autonomous robots

----- End of Sample Question Set -----

Q35. Explain why the way a project will be evaluated should be settled at the start.

- A) It reduces the cost of the computing hardware
- B) It decides what is built and what counts as success
- C) It removes the need to collect any data
- D) It guarantees the model will be accurate

Q36. Interpret a system described as reactive in the functionality classification.

- A) It remembers everything it has ever been shown
- B) It reasons about the beliefs of other agents
- C) It responds to the present input and remembers nothing
- D) It holds an internal model of its own state

Q37. Predict the effect of deploying a model with no plan for monitoring it.

- A) The model will stop running after a fixed period
- B) The model will improve steadily without help
- C) A gradual decline in performance will go unnoticed
- D) The model will retrain itself when it drifts

Q38. Describe what the privacy principle requires of a system that uses personal records.

- A) That the data is published for public inspection
- B) That only necessary data is collected and kept securely
- C) That every record is copied to a second server
- D) That the model is retrained every single week

Q39. Explain why generative Artificial Intelligence raises questions about authorship and consent.

- A) It cannot produce any text longer than a page
- B) It stores a copy of every question it is asked
- C) It requires a licence for every generated word
- D) It is trained on work produced by identifiable people

Q40. Distinguish an agent's environment from its architecture.

- A) The environment is the world, the architecture the machinery
- B) The architecture is the world, the environment the machinery
- C) Both of the two describe the surroundings of the agent
- D) Both of the two describe the hardware of the agent